

Universidad Nacional Abierta y a Distancia
Vicerrectoría Académica y de Investigación
Course: Natural Language Processing
Code: 203238430

Learning Guide - Phase 2 - Fundamentals of Natural Language Processing

1. Information of the Phase

Table 1. *Description.*

Aspect	Description
1. Type of activity	Independent
2. Evaluation moment	Intermediate
3. Managing unit	Escuela de Ciencias Básicas Tecnología e Ingeniería ECBTI
4. Score of the Choose an option	120 points
5. The activity starts on:	Monday, August 31, 2026
6. The activity ends on:	Saturday, September 26, 2026
7. Student independent work hours	32 hours

2. Detailed Description of the Learning Activity

Through this activity, the following learning outcome is expected to be achieved:

Learning Outcome 1: Adapt lexical resources and text corpora for text analysis and develop structured programs with Python for raw text.

This activity is designed to consolidate core competencies in Natural Language Processing (NLP) at the master's level. By the end of the phase, the student will be able to apply lexical resources and text corpora for text analysis and develop structured Python programs to extract relevant information from plain text and categorize it using NLP techniques.

The activity uses the CISI corpus ("CISI: a dataset for Information Retrieval"), a public benchmark collection from the University of Glasgow's Information Retrieval Group. The dataset is freely available on Kaggle, **Annex 1 - dataset for Information Retrieval.zip**.

The CISI collection contains 1,460 documents (mainly scientific and technical abstracts in library and information science), 112 queries and relevance assessments that map each query to its relevant documents. This corpus will serve as the common reference dataset for all the exercises in this phase.

The activity focuses on the following general outcomes:

- Developing an NLP pipeline that incorporates fundamental text preprocessing techniques.
- Applying supervised machine learning classification techniques and evaluating them with standard performance metrics.
- Synthesizing the methodological process and results in a technical, article-style report aligned with graduate academic standards.

To successfully complete this activity, students must have access to:

- The CISI corpus ("CISI: a dataset for Information Retrieval") downloaded from Kaggle (University of Glasgow Information Retrieval Group), in a convenient text format (e.g., the CISI.ALL, CISI.QRY and CISI.REL files).
- A Python development environment (such as Jupyter Notebook, Google Colab, or a similar tool).
- Libraries for NLP and machine learning (for example: NLTK, spaCy, scikit-learn, gensim, and relevant deep learning frameworks).

Exercise 1: Text Preprocessing Techniques

Description of the exercise:

Using the CISI corpus as the input dataset, implement, in a well-documented Python Notebook, a complete text preprocessing pipeline. Perform cleaning and normalization of the text by incorporating the following techniques:

- Tokenization.
- Stopword removal.
- Case normalization (conversion to lowercase).
- Removal or normalization of jargon (slang), abbreviations, and diacritics (accents).
- Noise removal (e.g., punctuation, URLs, numbers, or other non-informative tokens, as appropriate).

- Stemming.
- Lemmatization.

Subsequently, implement and compare the following text representation techniques:

- Bag of Words (BoW).
- Bigrams (and optionally higher-order n-grams, if justified).
- Term Frequency-Inverse Document Frequency (TF-IDF), considering each document as a unit of analysis.

Go a step further by incorporating distributed representations to obtain dense information vectors, such as:

- Word2Vec.
- Global Vectors for Word Representation (GloVe).

Prepare a comparative analysis table that includes each preprocessing and representation technique, highlighting its advantages, limitations, implementation challenges, and any relevant observations. Each technique must be analyzed explicitly and critically.

Exercise 2: Model Training and Metrics Assessment

Description of the exercise:

Using the processed CISI corpus, train and evaluate a set of supervised machine learning models for text classification. At least, implement the following algorithms:

- Logistic Regression.
- Naïve Bayes.
- k-Nearest Neighbors (k-NN).
- Decision Trees.
- Support Vector Machines (SVM).
- Random Forests.
- An ensemble method (Boosting or Bagging).
- A Neural Network model (architecture of your choice, properly justified).

Evaluate and compare the models using standard performance metrics. For classification problems, report at least: accuracy, confusion matrix,

precision, recall, and F1-score. For regression tasks (if applicable), report: R^2 , mean squared error (MSE), and any other relevant metric.

Document an intermediate technical report in which you summarize the experimental design, hyperparameter choices, and the comparative performance of the models. Clearly justify the selection of evaluation metrics and any model tuning decisions.

Exercise 3: Presentation of activity report

Description of the exercise:

Prepare a final technical report in article format that consolidates the outcomes of Exercises 1 and 2. The report should be written in clear, formal academic English and include the following sections:

- Abstract: A concise summary of the objective, methodology, main results, and conclusions.
- Introduction (optional): Contextualization of the problem, relevance of NLP and information retrieval for the CISI corpus, and objectives of the activity.

Methodology:

- Corpus description: number and type of attributes, number of instances, and data source (CISI dataset from the University of Glasgow, accessed via Kaggle).
- Data preprocessing and feature selection or extraction.
- Models and optimization procedures (including hyperparameter search strategies).
- Experimental design and validation strategy (e.g., train-test split, cross-validation).

Results:

- Performance metrics for each model.
- Comparative tables and, when appropriate, visualizations summarizing the main findings.

Discussion and Conclusions:

- Identification of the best- and worst-performing models and analysis of their behavior.
- Interpretation of any unexpected or striking results, supported by theory or empirical evidence.
- Assessment of whether the best-performing model is suitable for deployment in an organizational context and under which conditions.
- Reflections on the limitations of the study and suggestions for future improvements or extensions.

All individual or group written products must demonstrate rigorous academic writing, including appropriate structure, coherence, cohesion, and formal style consistent with master's-level work.

References must follow the current APA standards. Students are strongly encouraged to use the Turnitin tool available on the virtual campus to review and improve the originality and academic integrity of their written work before submission.

3. Academic Integrity and Institutional Regulations

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University"

The academic sanctions that the student will face are as follows:

- a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.
- b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.